BALKIS SELMI

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SUMMARY

Geomatics Engineering student experienced in GIS, remote sensing, Python automation, and AI-driven analysis. Contributed to seismic data processing at ETAP and solar PV forecasting at STEG. Interested in applying geospatial technologies and machine learning to environmental monitoring, climate resilience, and renewable energy.

PROFESSIONAL EXPERIENCE

Intern – Seismic Data Processing , Entreprise Tunisienne d'Activites Petrolieres (ETAP), Tunisia , June 2026

- Automated seismic receiver assignment for 100+ receivers across 50 shot points, reducing manual processing time by ~40% through ArcGIS Pro 3 toolbox development
- Engineered Z-value interpolation algorithms that improved elevation data accuracy by 15%, eliminating field rework costs
- Streamlined petroleum exploration workflows by integrating geospatial automation, cutting data processing bottlenecks for seismic survey teams

Data Science & AI Intern, STEG, June 2026

- Architected centralized monitoring dashboard aggregating real-time operational data from 53+ district solar PV stations, replacing fragmented spreadsheets and reducing repo rting time by 60%
- Integrated production metrics, power generation logs, and inverter fault detection into unified interface, enabling faster fault resolution and minimizing downtime across the solar fleet
- Developed machine learning model achieving 85%+ forecast accuracy for photovoltaic energy production 3 months in advance, optimizing grid integration and maintenance scheduling

Data Annotator, Doxaria, September 2025 - Present

- Labeled and categorized 500+ medical documents with 10,000+ precise annotations, achieving 98% quality assurance scores to train high-performance AI/ML models
- Directed team of 8 annotators to deliver 100% of projects on deadline, implementing quality control protocols that reduced error rates by 25%
- Verified data accuracy against clinical guidelines, ensuring zero critical misclassifications in sensitive medical datasets

Intern - Geomatics & Remote Sensing, Entreprise Tunisienne d'Activites Petrolieres (ETAP), Tunisia, July 2025

- Processed geospatial data and performed coordinate system transformations using Petrel and ArcGIS Pro.
- Contributed to documentation and analysis of mapping systems (IGN and RGS) used in Tunisia.
- Gained practical experience in applying geodesy and geomatics in the oil and gas sector.

Job Simulation - Engineer for New Energy, APA Group, Australia, February 2025

- Conducted a professional simulation within the Project Development team on renewable energies.
- Applied concepts related to solar photovoltaics and battery storage to draft a Basis of Design.
- Calculated installed capacity, load factor, and potential CO2 emissions reduction of a solar power plant.
- Developed awareness of safety aspects and gained hands-on experience in troubleshooting and problem-solving during operations.

Intern - Python Developer, Code Alpha, India, August 2024 - September 2024

- Developed and optimized 3 Python applications, improving script execution speed by 35% through algorithm refinement
- Designed 2 desktop applications using PyQt5 with intuitive GUIs, reducing manual data entry time for end users by 50%
- Collaborated on innovative solutions adopted by the development team, strengthening technical architecture standards

LEADERSHIP, CIVIC ACTIVITIES & TEACHING

Expert member, BLUE MSE, September 2026 – Present

- Contribute technical and strategic expertise to environmental and sustainability projects.
- Support the development of initiatives focused on climate action, innovation, and sustainable development.
- Collaborate with members on project design, implementation, and knowledge sharing.

Member, JCI Belvédère Tunisia, January 2025 – Present

- Participate in civic engagement, leadership, and community development initiatives.
- Contribute to activities focused on youth empowerment, social impact, and community involvement.

General Secretary, BLUE MSE, September 2025 – September 2026

- Lead the association's administrative coordination and internal communication.
- Coordinate 2 hackathons, more than 5 environmental and sustainability initiatives (at regional scale), around 5 webinars, and student-led projects.
- Manage communication and collaboration between board members, members, partners, and external stakeholders.
- Organize meetings, documentation, and follow-up of ongoing activities.
- Contribute to projects promoting climate action, environmental awareness, and sustainable development.

Arabic Tutor, Preply, March 2025 - Present

- Delivered personalized 1:1 Arabic (MSA and conversation) lessons tailored to students' goals and proficiency levels.

- Designed lesson plans covering speaking, grammar, vocabulary, reading, and pronunciation.
- Monitored student progress and adapted teaching strategies to improve fluency and confidence.

Social Media & Community Manager, Association Tunisienne des Educateurs, June 2024 - Present
- Created and published engaging content promoting educational initiatives and events.
- Managed the association's online presence and digital brand image.
- Coordinated events, workshops, partners, and communication campaigns to increase visibility and participation.

EDUCATION

Engineering Degree in Geomatics, Manouba School of Engineering, August 2023 - Present
- Preparatory cycle: Sep 2023 - June 2025.
- Engineering cycle: Aug 2025 – June 2028. degree expected June 2028.

Secondary School Diploma, Pionner High School of Ariana, September 2019 - June 2023
- Distinction: Very Good - Score: 17.28/20.

PROJECTS

- Research: 'A new framework to transform education in Tunisia: SDGs as a lever for progress' - ENIM (December 2024).
- Developed and implemented a coordinate system conversion in Petrel (Carthage RGS to Carthage IGN) (July 2025).
- HydroTwin - Intelligent water management solution (July 2025 - Present).
- Drought Monitoring Dashboard - Interactive GIS dashboard using satellite imagery and Python to monitor drought conditions (December 2025 - May 2026).
- HydroTwin Mobile App - Digital twin dashboard integrating GIS and AI to predict flooding and water risk zones (December 2025).
- Aquora - AI- and GIS-based virtual living lab for water monitoring, water security, and flood-risk assessment (June 2026 - Present).

TECHNICAL SKILLS

- GIS & Remote Sensing: ArcGIS Pro, QGIS, ENVI, Google Earth Engine, Georeferencing, Raster Analysis, Vector Analysis, Spatial Modeling, Environmental Change Detection, Geostatistics.
- Programming & Data Science: Python, NumPy, Pandas, Matplotlib, SQL, PostgreSQL, C++, MATLAB, Data Cleaning, Data Preprocessing, Feature Extraction, Model Training, Model Evaluation.
- AI & Machine Learning: Supervised Classification, Unsupervised Classification, Clustering, Neural Network Fundamentals, Data Labeling, AI-based Remote Sensing Applications.
- Image Processing: Image Segmentation, Object Detection, Edge Detection, Histogram Equalization, Morphological Operations, Filtering, Image Enhancement.
- Web & Mobile Development: HTML, CSS, JavaScript, React, Django, PHP, Android Studio.
- Other Tools & Platforms: Petrel, PyQt5, Excel, Jupyter Notebook, Google Colab, GitHub.

CERTIFICATIONS

- Green Digital Certificate Program - INCO Academy in partnership with LinkedIn (September 2025).
- Engineering for New Energy Job Simulation - APA Group, via Forage (February 2025).
- IBM TechXchange Dev Day: Virtual Agents - Built a virtual agent in a hands-on developer workshop (January 2025).
- Diploma in C Programming - Alison, final assessment score 88% (January 2025).
- Six Sigma Yellow Belt - 6sigmastudy, Certificate ID 901712 (January 2025).
- Python Programming Virtual Internship - CodeAlpha (September 2024).

LANGUAGES

- Arabic, English, French, Spanish

SOFT SKILLS

- Analytical Thinking, Attention to Detail, Communication, Collaboration, Problem-Solving, Adaptability

AWARDS

- Top 10 National Winner - Represented Tunisia at the international finals of Arab Reading Challenge in Dubai (2016).
- National Second Prize - Filmmaking Production, 'Creatives in the Quarantine' competition (2021).
- Presented the HydroTwin project at the International Space Star Conference and received certification for the scientific poster presentation (2025).
- Selected among the Top 15 national research ideas in BrainX; completed the BrainX Bootcamp organized by Open Startup and received certification (2025).
- Selected by JCI TBS Tunisia for the TOYP 2026 (Ten Outstanding Young Persons) program and advanced to the Top 30 national finalists for the African stage.
- Selected as a delegate for the RCOY MENA 2026 program - among 500 delegates selected from 18,000 participants.
- Participated in Oman Climate Week 2026, engaging in climate-related discussions and activities in Oman.